

POWERFACTORY

PowerFactory 2024

Technical Reference

Thyristor-Controlled Series Compensation

ElmTcsc

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

PowerFactory provides an AC thyristor-controlled series compensation (TCSC) model.

2 Basic Data

Figure 2.1 shows the basic scheme of the TCSC.

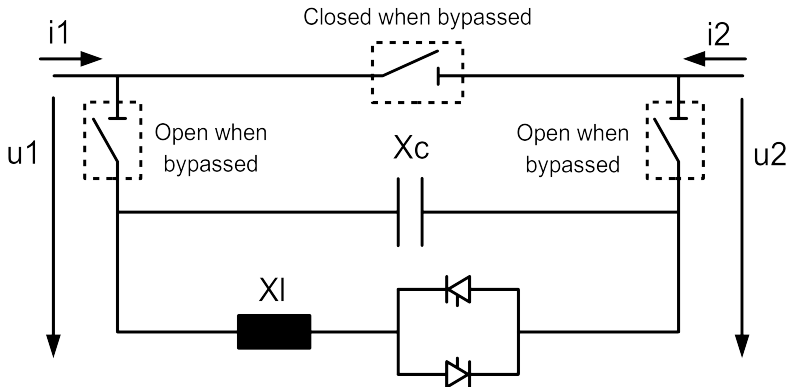


Figure 2.1: Basic Scheme

If the option *Bypassed* is enabled the corresponding bypass switch is closed and the series capacitor is isolated by opening the corresponding separators.

3 Load Flow Analysis

Typically, in load flow calculations, the controlled operation of a TCSC is modelled. The following operation modes are supported:

- Active power: see Section 3.1.
- Reactive power: see Section 3.1.
- Current magnitude: Controls the magnitude of the current value going through the TCSC.
- Effective reactance: Controls the effective reactance of the TCSC.
- Transmission angle: Controls the difference of the voltage angle at both terminals (*u:phiu:1 - u:phiu:2*).

For each of these methods, the effective reactance (X_{TCSC}) in Ω is calculated so that following equations are satisfied:

$$\underline{u_1} - \underline{u_2} = \underline{i_1} \cdot \frac{(jX_{TCSC})}{U^2_{nom:1}} \quad (1)$$

$$\underline{i_1} + \underline{i_2} = 0 \quad (2)$$

$$actualvalue = setpoint \quad (3)$$

The values of voltage and current are in p.u. based on 1 MVA and $U_{nom:1}$.

From the effective reactance, the firing angle (α) is calculated, such that the equation 4 is satisfied.

$$\begin{aligned} X_{TCSC}(\alpha) = & - X_C + C_1\{2(\pi-\alpha) + sin[2(\pi-\alpha)]\} \\ & - C_2 cos^2(\pi-\alpha)\{k_x tan[k_x(\pi-\alpha)] - tan(\pi-\alpha)\} \end{aligned} \tag{4}$$

Where:

- Xc is the capacitor reactance
- Xl is the thyristor-controlled reactance
- $k_x = \sqrt{\frac{Xc}{Xl}}$
- $X_{LC} = \frac{Xc \cdot Xl}{Xc - Xl}$
- $C_1 = \frac{Xc + X_{LC}}{\pi}$
- $C_2 = \frac{4X_{LC}^2}{\pi Xl}$

3.1 Power control

The active or reactive power can be controlled at the local terminal (terminal i or terminal j can be selected), or at a remote location (remote cubicle or boundary).

When the remote location has been selected, an orientation parameter *iOrient* has to be specified. The options are the following:

- +P, +Q: At these locations, any increase in the measured power will result in more power flowing through the TCSC, which in turn will decrease the effective reactance.
- -P, -Q: At these locations, any increase in the measured power will result in less power flowing through the TCSC, which in turn will increase the effective reactance.

The figure 3.1 explains it graphically.

3.2 Limits

The limits can be specified as minimum and maximum effective reactance values. The firing angle is then calculated from the computed effective reactance during the load flow solution, within a range of 90° to 180°. If there is at least one pole within this range, then the first firing angle found will be displayed. If no firing angle value was found, then the displayed value is zero.

Alternatively, the limits can be given as minimum and maximum firing angles.In this case, the respective effective reactance limits will be calculated using equation 4. If there is at least one

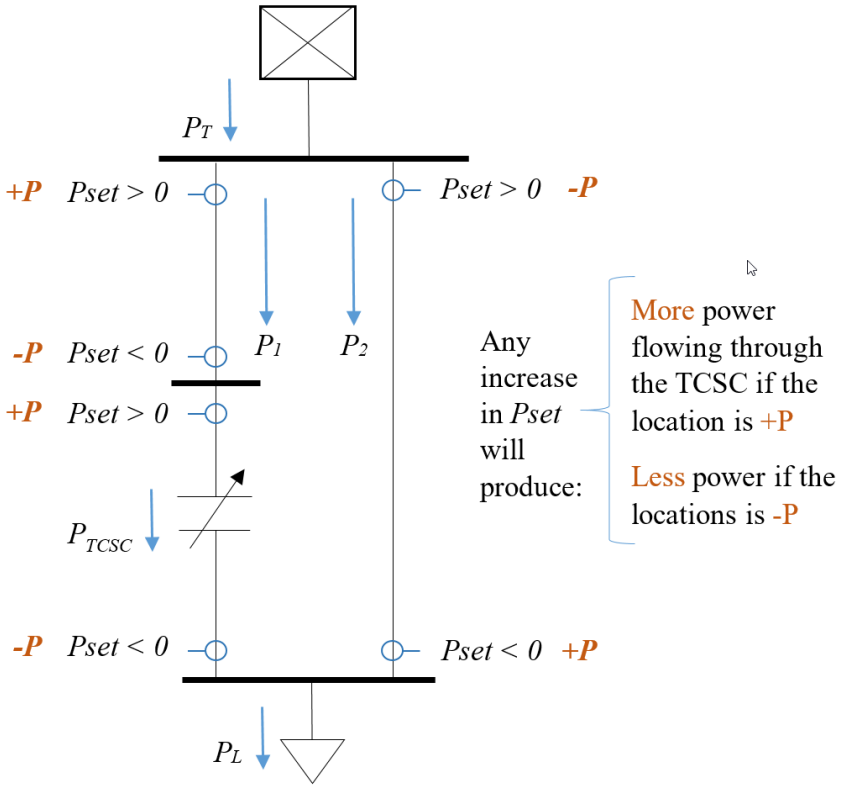


Figure 3.1: Orientation parameter

pole within this range, then the effective reactance limits will not be considered and the first firing angle found will be displayed. If no firing angle value was found, then the displayed value is zero.

The number of poles in the specified range are calculated by equation 5.

$$\alpha_{TCSC} = \pi \left[1 - \frac{(2n-1)\omega\sqrt{LC}}{2} \right], n = 1,2,3... \tag{5}$$

In addition, the maximum voltage drop/rise can be specified. This parameter modifies the effective reactance, so that the voltage drop/rise does not exceed the given value. The modified effective reactance will still respect the maximum/minimum reactance limits.

3.3 Quasi-Dynamic Simulation

For the quasi-dynamic simulation, the following input signal is available:

- XTCSCin, effective reactance in Ω

When this signal is connected, the value is used directly in Equation 1, and disregarding Equation 3, thus disabling the specified control mode.

3.4 Calculation Quantities

Loading

The loading of the TCSC is calculated as follows:

$$loading = \frac{max(I_{bus1}, I_{bus2})}{I_{nom}} \cdot 100\ \%$$

- $loading$: Loading in %
- I_{nom} : Nominal current $= I_r$ of the TCSC in kA
- I_{bus1} : Magnitude of the current at terminal i
- I_{bus2} : Magnitude of the current at terminal j

If a thermal ratings object is selected, the nominal current I_{nom} is determined as follows:

- if the continuous rating is entered in MVA:
  - 3-phase TCSC:

    $$I_{nom} = ContRating/(\sqrt{3} \cdot U_{n(bus1)})$$

  where $U_{n(bus1)}$ is the nominal voltage in kV of the connected terminal i
- if the continuous rating is entered in kA:

  $$I_{nom} = ContRating$$

- if the continuous rating is entered in %:

  $$I_{nom} = ContRating/100 \cdot I_r$$

Further details regarding thermal ratings object (*IntThrating*), see Technical Reference of the Thermal Ratings Definition

For an unbalanced load flow calculation the highest current of all phases is used.

Losses

The losses are calculated as follows:

Quantity	Unit	Description	Value
$Ploss$	MW	Losses (total)	$= 0$
$Qloss$	Mvar	Reactive-Losses (total)	$= 0$
$Plossld$	MW	Losses (load)	$= 0$
$Qlossld$	Mvar	Reactive-Losses (load)	$= 0$
$Plossnld$	MW	Losses (no load)	$= 0$
$Qlossnld$	Mvar	Reactive-Losses (no load)	$= 0$

Table 3.1: Losses Quantities, AC-model

Voltage Drop

Quantity	Unit	Description	Value
du	p.u.	Voltage Drop	$= \lvert\underline{u}_{bus1}\rvert - \lvert\underline{u}_{bus2}\rvert$
$dupc$	%	Voltage Drop	$= du \cdot 100$
$dphiu$	deg	Voltage Drop Angle	$= \phi_{u,bus1} - \phi_{u,bus2}$
$du1$	p.u.	Positive Sequence Voltage Drop	$= \lvert\underline{u1}_{bus1}\rvert - \lvert\underline{u1}_{bus2}\rvert$
$du1pc$	%	Positive Sequence Voltage Drop	$= du1 \cdot 100$
$dphiu1$	deg	Positive Sequence Voltage Drop Angle	$= \phi_{u1,bus1} - \phi_{u1,bus2}$

Table 3.2: Voltage Drop Quantities, AC-model

u_{bus1} and u_{bus2} are the corresponding terminal voltage in p.u. based on the rated voltage of the terminal. $\phi_{u,bus1}$ and $\phi_{u,bus2}$ the terminal voltage angle in deg. For an unbalanced load flow du, $dupc$, $dphiu$ is per phase available, e.g. $c : dupc : B$.

3.5 DC Load Flow (linear)

For the DC load flow, only the following control methods are supported:

- Active power: see Section 3.1.
- Effective reactance: Controls the effective reactance of the TCSC.

For each of these methods, the effective reactance (X_{TCSC}) in Ω, is calculated so that the followings equations are satisfied:

$$(\phi_{bus1} - \phi_{bus2})/X_{rea} = P_{bus1} \tag{6}$$

$$P_{bus1} + P_{bus2} = 0 \tag{7}$$

where ϕ_{bus1} is the voltage angle on terminal i, ϕ_{bus2} angle on terminal j.

4 Short-Circuit Analysis

4.1 VDE/IEC Short-Circuit

Thyristor-controlled series compensation models use the effective reactance, calculated using equation 4 and assuming a firing angle of $\alpha = 90°$, during VDE/IEC short-circuit calculations.

During R/X ratio calculations, or if the option *Bypassed* is enabled, the effective reactance is assumed to be zero.

4.2 Complete Short-Circuit

Thyristor-controlled series compensation models use the effective reactance during complete short-circuit calculations. If *Load Flow Initialisation* is enabled, the effective reactance resulting from the load flow calculation is used, otherwise it is calculated using equation 4 and assuming a firing angle of $\alpha = 90°$.

During R/X ratio calculations, or if the option *Bypassed* is enabled, the effective reactance is assumed to be zero.

The option *Consider bypass circuit* allows for integrated protective circuits (such as a spark gap or metal oxide varistor) to be considered automatically during the short-circuit calculation. Voltage and current thresholds for activating the bypass can be enabled separately or together. The phase current is compared with the current threshold (*e:shcIprot*) and the voltage across the TCSC with the voltage threshold (*e:shcVprot*).

If at least one threshold is enabled, the short-circuit calculation is carried out as follows:

1. Calculate the subtransient short-circuit without bypass;
2. Evaluate current and/or voltage against the thresholds and activate the bypass if necessary;
3. If the bypass is activated, the subtransient short-circuit is re-calculated.

The state of the bypass determined for the subtransient calculation is retained for the transient calculation.

For unbalanced short-circuit calculations, it is sufficient for one phase to exceed the threshold to trigger the bypass for the whole TCSC.

4.3 ANSI Short-Circuit

Thyristor-controlled series compensation models use the effective reactance, calculated using equation 4 and assuming a firing angle of $\alpha = 90°$, during VDE/IEC short-circuit calculations.

If the short-circuit command setting *Bypass Series Capacitor* is configured to a value other than "No Bypassing", TCSCs will be neglected (i.e. bypassed) for the selected calculations.

Depending on the calculation method used for the X/R calculation, the following applies:

Separate R and X TCSCs are always neglected (i.e. bypassed);

Complex TCSCs are considered according to the selection made for *Bypass Series Capacitor*.

5 RMS-Simulation

Two input signals are available:

- gatea, firing angle in deg
- XTCSC, effective reactance in Ω

The signal *gatea* has priority over *XTCSC*. When the *gatea* is used, the effective reactance is calculated using 4 and then using equations 1 and 2.

6 EMT Simulation

The EMT model is described by the following equations and the Figure 6.1.

Two input signals are available:

- gatea, firing angle in deg
- XTCSC, effective reactance in Ω

The signal *gatea* has priority over *XTCSC*. When the *XTCSC* is used, the firing angle is calculated using 4 and then using the following equations:

$$iC_A = c \cdot \frac{du_{C:A}}{dt} \tag{8}$$

$$\frac{diL_A}{dt} = \begin{cases} 0 & \text{if thyristor off} \\ \frac{u_{C:A}}{l} & \text{if thyristor on} \end{cases} \tag{9}$$

$$i_{1:A} = iC_A + iL_A \tag{10}$$

$$i_{1:A} + i_{2:A} = 0 \tag{11}$$

The equations are the same for phases B and C.
The values of voltage and current are in p.u. based on 1 MVA and $U_{1:A}$.

In addition the following currents and voltage are available in kA or kV:

- $Ucap:A$,$Ucap:B$ and $Ucap:C$: Capacitor voltage in kV
- $Icap:A$,$Icap:B$ and $Icap:C$": Current through Capacitor in kA
- $Itcr:A$,$Itcr:B$ and $Itcr:C$,: TCR Current in kA
- $Imov:A$,$Imov:B$ and $Imov:C$,: MOV Current in kA
- $Ibybrk:A$,$Ibybrk:B$ and $Ibybrk:C$,: Bypass Breaker Current in kA
- $Ubybrk:A$,$Ubybrk:B$ and $Ubybrk:C$,: Bypass Breaker Voltage in kV

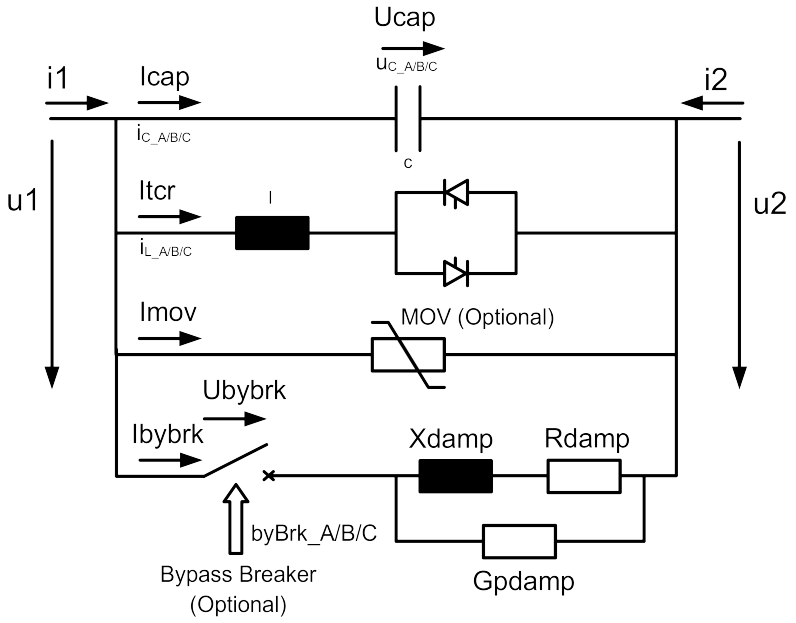


Figure 6.1: EMT Model

6.0.1 With Metal Oxide Varistor Protection

When the option *Metal Oxide Varistor* is selected, a non linear resistance can be defined which represents the protection of the TCSC by the MOV. The non linear resistance is entered by a voltage-current characteristic table. The entire characteristic is then approximated using cubic splines or alternative with a piecewise linear interpolation.

The model equations of the TCSC with MOV protection are:

$$i_{MOV}(t) = f(u_{12}(t)) \tag{12}$$

The dissipated MOV energy is calculated as follows:

$$P_{MOV}(t) = i_{MOV}(t) \cdot u_{MOV}(t) \tag{13}$$

$$E_{MOV}(t) = \int_{tstart}^{t} P_{MOV}(\tau) d\tau \tag{14}$$

6.0.2 Bypass Breaker

The TCSC model for EMT simulations allows the definition of a protection bypass breaker, via the selection of the *Bypass Breaker* option. Once this option is selected, the resistance,reactance and the parallel conductance of the damping circuit can be defined. The bypass breaker model with optional parallel MOV surge arrester is shown in Figure 6.1.

External Tripping

When the option *Built-In Tripping Unit* is deactivated, the bypass breaker can be triggered via the use of input signals *byBrk_A*, *byBrk_B* and *byBrk_C* (defined in Table A.1) using either a parameter event (*EvtParam*) defined for the EMT simulation, or via an external DSL controller. The surge arrester is described in detail in the *PowerFactory* Surge Arrester Technical Reference document.

Built-In Tripping Unit

By activating the option *Built-In Tripping Unit*, it is possible to specify when to trigger the bypass breaker, according to any of the following thresholds:

- Trip when Current value is above the specified value in kA
- Trip when Voltage value is above the specified value in kV. Available only for option *Metal Oxide Varistor* deactivated.
- Trip when Energy value is above the specified value in MW. Available only for option *Metal Oxide Varistor* selected.
- Trip when Power value is above the specified value in MW. Available only for option *Metal Oxide Varistor* selected.

Concurrent Tripping

The triggering of the models can be done concurrently by the built-in tripping conditions as well as by using the external signals.

A Signals Definitions

Table A.1: Internal signals (EMT-Simulation)

Variable	Unit	I/O	Description
gatea	deg	IN	Effective reactance
XTCSC	Ω	IN	Firing Angle
Fmeas	Hz	IN	Frequency
iL_A	p.u.	OUT	TCR Current, A
iL_B	p.u.	OUT	TCR Current, B
iL_C	p.u.	OUT	TCR Current, C
iC_A	p.u.	OUT	Capacitor current, A
iC_B	p.u.	OUT	Capacitor current, B
iC_C	p.u.	OUT	Capacitor current, C
phi_A	rad	STATE	Voltage Angle, Phase A
phi_B	rad	STATE	Voltage Angle, Phase B
phi_C	rad	STATE	Voltage Angle, Phase C
Emov_a	MWs	STATE	MOV absorbed energy, A
Emov_b	MWs	STATE	MOV absorbed energy, B
Emov_c	MWs	STATE	MOV absorbed energy, C
byBrk_A		IN	Bypass Breaker, A
byBrk_B		IN	Bypass Breaker, B
byBrk_C		IN	Bypass Breaker, C

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